



SEMQ: Efficient non-uniform quantization with sensitivity-based error minimization for large language models

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ABSTRACT

Large Language Models (LLMs) represent a pivotal breakthrough in computational intelligence, showcasing exceptional capabilities in information aggregation and reasoning. However, their remarkable performance comes at the cost of ultra-high-scale parameters, leading to significant resource demands during deployment. Therefore, various model compression techniques have been developed, such as pruning, distillation, and quantization. Among these, quantization has gained prominence due to its ability to directly reduce the precision of model weights and activations, resulting in substantial memory savings and accelerated inference. Despite its advantages, existing quantization approaches face substantial challenges in ultra-low precisions (e.g., 2-bit), often resulting in severe performance degradation. To tackle this challenge, we propose a novel non-uniform quantization with minimal disturbance for LLM, which contains two innovations: (i) a Sensitivity-based Error Minimization Non-Uniform Quantization (SEMQ) algorithm, which finds the quantization scheme to minimize the quantization error through continuous iteration; and (ii) a Z-score-based method for outlier detection and isolation under the normal distribution assumption, reducing the complexity of the quantization process. The extensive experiments on the LLaMA family demonstrates that the proposed SEMQ enables the ultra-low precision quantization up to 2-bit, and 10× GPU memory reduction for origin LLMs while maintaining the model accuracy. Our code is publicly available at <https://github.com/ldm2060/semq>.

1. Introduction

In recent years, the rapid development of Large Language Models (LLMs) has profoundly transformed various fields, marking a significant technological revolution in computational intelligence. Models such as the LLaMA family [1,2] (with parameters ranging from 7B to 65B), Gemma [3,4] (ranging from 2B to 27B) and GPT-4 [5] (estimated at 1.8 trillion parameters) have demonstrated unprecedented capabilities in natural language understanding, generation, and complex reasoning tasks. These advancements have not only expanded the frontiers of machine learning, but also enabled groundbreaking applications across a wide range of domains.

However, the remarkable performance of LLMs comes at a significant cost: their massive scale demands substantial computational and memory resources, making deployment challenging, especially in resource-constrained environments. For instance, the LLaMA-65B model requires

at least 130 GiB of memory to be deployed with FP16 precision, far exceeding the capacity of current consumer GPUs. Even smaller models like LLaMA2-7B demand 28 GiB of memory, surpassing the 24 GiB capacity of the high-end consumer GPU (NVIDIA RTX 4090). These resource requirements not only increase deployment costs, but also hinder the deployability of LLMs on resource-constrained devices, such as personal computers or smartphones. To address these challenges, there is a growing need for effective model compression techniques that can reduce resource consumption while significantly retaining the performance of LLMs, enabling their widespread adoption in practical scenarios [6].

Current approaches for LLM compression primarily include pruning, knowledge distillation, and quantization [7]. In detail, pruning reduces computation and resource usage by removing unimportant weights in the model, which involves identifying and eliminating redundant connections that do not significantly impact performance. Distillation

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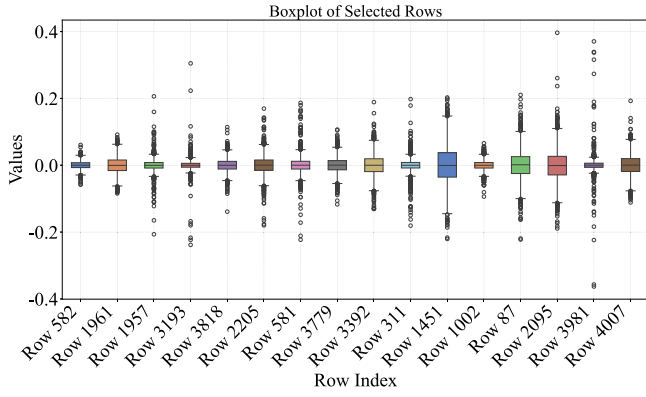


Fig. 1. The illustration of linear weight values of the LLaMA-7B, visualized by box plots. we randomly selected 16 rows from the weight matrix of the first layer in the LLaMA-7B model. According to the definition of the box plot, the black line in the middle of the box represents the median weight, the box boundaries represent the first (lower) and third (upper) quartiles, and the circles on the upper and lower sides of the box represent outliers outside the quartile.

transfers the knowledge of larger LLMs (referred to as the teacher model) to a smaller pre-designed model (the student model), allowing the smaller model to maintain comparable performance while consuming fewer resources. Unlike pruning and distillation, which often require extensive retraining or architectural modifications, quantization offers a hardware-friendly and deployment-efficient solution by directly reducing the precision of weights and activations, which has made it a central focus of both research and practical applications in model compression. Quantization can be divided into quantization-aware training (QAT) and post-training quantization (PTQ), depending on whether quantization is integrated during or after model training [8]. QAT introduces pseudo-quantization modules during training to reduce quantization errors, while PTQ applies quantization as a downstream process and relies on calibration techniques to maintain accuracy. Recently, quantization has become increasingly popular for its ability to significantly reduce model size and computational cost. Notably, PTQ offers greater practicality by compressing pre-trained models without requiring retraining, making it a scalable and efficient solution for resource-constrained deployment.

1.1. Motivation

Despite the success of post-training quantization (PTQ) in compressing models, quantizing Large Language Models (LLMs) presents significant challenges. Two main difficulties arise when reducing quantization errors while compressing LLMs. First, selecting an appropriate quantization method is a critical challenge. As illustrated in Fig. 1, the box plot depicts 16 randomly selected rows of weight values from the first layer of the LLaMA-7B model, with each row containing 4096 weights. The majority of the weights are tightly clustered around the center of the distribution, indicating a high degree of concentration. However, a few outliers can be observed, deviating significantly from the central tendency. Currently, most quantization methods rely on uniform quantization, which evenly distributes quantization points across the weight range. However, this approach ignores the bell-shaped distribution of LLM weights, leading to suboptimal results. Therefore, how to design appropriate quantization methods that can effectively account for the bell-shaped weight distribution of LLMs is essential to minimize quantization errors and preserve model performance. Secondly, the presence of outliers and sensitive values complicates quantization. Outliers are weight values that are very different from most others, as shown in Fig. 1. During quantization, outliers necessitate the allocation of a disproportionate number of limited quantization levels, effectively wasting these valuable resources. This inefficiency reduces the overall accu-

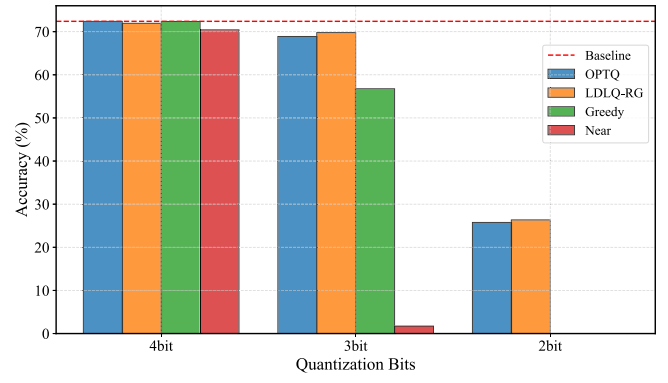


Fig. 2. The trend of the accuracy of some common quantization methods as the number of quantization bits decreases. RTN refers to the traditional Round-to-nearest quantization, LDLQ-RG and Greedy methods are two parts of QuIP [9]. The benchmark model is OPT-30B, and the evaluation dataset is LAMBADA [10]. The performance indicator for this dataset is accuracy, where a higher value indicates better performance. The red dotted line is the accuracy of the FP16 precision model. It can be found that when the numerical precision of the model is transferred to 2 bits, the performance of the model drops sharply.

racy of the quantized model, particularly when numerous outliers are present. Sensitive values, on the other hand, are weight values that significantly impact the model's performance even though they may not be extreme outliers. These values are harder to identify but can cause substantial errors if altered even slightly during quantization. Although the two kinds of parameters are important, current quantization methods cannot handle them well, causing difficulties in model quantization at low bit-widths.

Both of these issues can significantly impair the model's performance. To address these challenges, several quantization methods for large language models (LLMs) have emerged. For instance, OPTQ [11] (also known as GPTQ) optimizes the quantization error by approximating the diagonal of the Hessian matrix. AWQ [12] reduces post-quantization errors by identifying and retaining a portion of the sensitive weights. Additionally, SpQR [13] mitigates quantization difficulties by detecting outliers and storing them separately. These methods aim to enhance the accuracy and efficiency of quantized models while minimizing the impact of outliers and sensitive values. Despite achieving significant progress, these methods have failed to deliver good results in extreme cases such as 2-bit quantization. As shown in Fig. 2, OPTQ [11] can maintain full precision performance under 4 bit quantization. However, its effectiveness drops by more than 50 % under 2 bit quantization, and RTN and Greedy (a part of Quip [9]) almost become ineffective. This is because the limited precision exacerbates the impact of outliers and sensitive values, and general uniform quantization cannot select a better quantization correspondence in this case.

1.2. Main contributions

To address the aforementioned limitation, we propose an efficient quantization algorithm specifically designed for LLMs. Unlike conventional uniform quantization methods, which distribute quantization points evenly across the entire weight range, our approach employs a non-uniform quantization strategy. This strategy leverages non-linear relationships to allocate quantization points in a way that prioritizes high-frequency weights, which are concentrated near the mean of the bell-shaped distribution. By assigning more quantization points to these critical weights, our method effectively captures the inherent distribution characteristics of LLM weights, thereby significantly reducing quantization errors and improving overall model performance. On the other hand, to better handle outliers and sensitive values in the weights, we propose a novel outlier detection and separation mechanism to reduce the difficulty of subsequent weight quantization and avoid large

quantization errors after model quantization due to the existence of outliers, leading to a decrease in the accuracy of LLM after quantization. Then by integrating the non-uniform quantization strategy and the iterative method for minimizing quantization error based on model sensitivity, a novel non-uniform quantization with sensitivity-based error minimization (SEMQ) algorithm is proposed, enabling ultra-low precision model quantization up to 2 bits. In summary, the contributions of this paper can be summarized as follows:

- We propose a new non-uniform quantization algorithm for LLMs, which iteratively optimizes quantization point selection to reduce errors.
- We approximate LLM weight distributions as bell-shaped curves and introduce a mechanism to detect and separate outliers, improving quantization robustness.
- Our approach is validated across a range of benchmark datasets, where it consistently achieves 2-bit precision with minimal accuracy loss, highlighting its robustness and generalization capability.

2. Related work

In the following, we will conduct a comprehensive survey and critical analysis of the recent advances in LLM quantization. This analysis will be structured around two primary categories: uniform quantization and non-uniform quantization. These categories reflect different strategies for reducing the precision of LLM weights and activations, each accompanied by its own strengths and limitations. The methods grouped into these categories share a common objective: addressing and mitigating the challenges inherent to LLM quantization, as outlined in the Introduction. These challenges arise from the unique characteristics of LLMs, such as their massive scale, complex architectures, and sensitivity to numerical precision.

2.1. Uniform quantization of LLMs

Within the realm of uniform quantization, methods typically involve mapping floating-point weights to a fixed set of integer values with uniform intervals. Hence, it is also known as Round-To-Nearest (RTN) Quantization. While this approach offers simplicity and efficiency, it can lead to significant information loss, particularly in the presence of weight distributions with non-uniform characteristics. It follows the following formula:

$$x_q = \text{clamp}(\text{round}(\frac{x}{\delta}) + z, -2^{N-1}, 2^{N-1} - 1), \quad (1)$$

where x is the weight before quantization, x_q is the weight after quantization, δ is the quantization factor, z is the zero point, and N is the number of quantization bits. Although uniform quantization is conceptually straightforward, directly applying it to LLMs often leads to significant quantization errors, which necessitates the development of advanced optimization strategies. As a result, various refinements have been proposed to reduce these errors and improve the effectiveness of uniform quantization methods.

One critical aspect of uniform quantization optimization concerns the configuration of quantization groups. Traditional uniform quantization typically operates at the granularity of the entire weight matrix. However, recognizing the inherent structure of matrix operations within LLM inference, alternative approaches have been explored. GPT3.int8() [14] introduced vector-wise quantization, which leverages the observation that model inference primarily involves matrix multiplications. Given a matrix multiplication $A \times B^T$, where each row of A is multiplied by each row of B , quantizing each column separately, rather than applying uniform quantization to the entire matrix, has been found to better preserve model accuracy.

As previously discussed, weight sensitivity plays a pivotal role in determining the impact of individual weights during model inference.

However, accurately quantifying weight sensitivity presents a significant challenge. OPTQ [11] building upon the Optimal Brain Compression (OBC) [15] method, provides a principled approach to estimating weight sensitivity. Let $\text{Loss}(w)$ represent the model training loss function, where w denotes the model weights. Given that weight values undergo subtle alterations during quantization, let Δ represent the change in weight value. A Taylor series expansion of the quantized loss function is then performed, as depicted in the following formula:

$$\text{Loss}(w + \Delta) = \text{Loss}(w) + \left(\frac{\partial \text{Loss}}{\partial w} \right)^T \Delta + \frac{1}{2} \Delta^T H \Delta + \mathcal{O}(\Delta^3), \quad (2)$$

where H is the Hessian matrix of the loss function. It can be observed that a well-trained model typically converges to a local minimum in the loss landscape, so at these minima, the gradient of the loss function approaches zero (i.e. $\frac{\partial \text{Loss}}{\partial w} = 0$). And based on the aforementioned Taylor expansion, we can conclude as follows:

$$\text{Loss}(w + \Delta) - \text{Loss}(w) = \frac{1}{2} \Delta^T H \Delta, \quad (3)$$

which reveals that the discrepancy in loss before and after weight quantization is intrinsically linked to both the differential between the pre- and post-quantization weights and the Hessian matrix of the loss function. Obviously, the Hessian matrix, which encapsulates the second-order partial derivatives of the loss function, provides a measure of the loss landscape's curvature. Consequently, weights residing in regions of high curvature, deemed 'sensitive weights', exhibit a greater change in loss upon quantization compared to weights located in flatter regions, which underscores the critical role of the Hessian matrix in quantifying weight sensitivities. Therefore, we can formally define the sensitivity of individual weights through Formula Eq. (3).

Subsequently, the strategic handling of sensitive values and outliers has emerged as a critical consideration in the pursuit of effective LLM quantization. Activation-aware Weight Quantization (AWQ) [12] posited that the weights within linear layers exhibit varying degrees of importance. Specifically, they identified a subset of significant weights, comprising approximately 0.1 % to 1 % of the total, that exert a disproportionate influence on model accuracy. AWQ advocates for preserving the original precision of these critical weights while quantizing the remaining weights, thereby maintaining overall model performance. Similarly, GPT3.int8() [14] recognized the presence of outliers, termed 'emergent features', which can substantially exacerbate quantization errors. To mitigate this issue, they proposed a mixed-precision decomposition method, wherein outliers are quantized with 16-bit precision, effectively reducing the overall quantization error. Extending this concept, SpQR [13] specifically targets the approximately 1 % of outliers, employing a sparse matrix representation after their identification and separation. This approach not only mitigates the adverse impact of outliers on model quantization but also preserves model accuracy through accurate representation of extreme values.

Beyond targeted outlier and sensitive value management, preprocessing techniques aimed at facilitating quantization by transforming weight distributions have also garnered significant attention. SmoothQuant [16] exemplifies this approach, employing a preprocessing methodology to transform activation layer weights and linear layer weights, rendering them more amenable to quantization while preserving model output invariance. AffineQuant [17] operates on a similar principle, utilizing scaling factors derived from weight distributions to construct an affine transformation matrix. OmniQuant [18] advances this concept by introducing a learnable equivalent transformation (LET), thereby enhancing the adaptivity and effectiveness of the preprocessing transformation.

2.2. Non-uniform quantization of LLMs

Non-uniform quantization represents a class of quantization techniques wherein the mapping between pre- and post-quantization weights is governed by a nonlinear function. While inherently more

complex than uniform quantization, non-uniform methods offer the potential for enhanced quantization accuracy by leveraging the structural characteristics of the model. Common instantiations of non-uniform quantization include techniques that employ nonlinear mapping functions or clustering algorithms, such as K-means [19], for weight discretization. nuLSQ [20] is a non-uniform quantization method that adapts to the long-tail distribution of weights and activations by optimizing multiple step sizes. Compared to traditional uniform quantization methods, nuLSQ can better preserve the original data distribution, thereby enhancing the model's expressiveness and performance. To improve the accuracy of quantized neural networks by bridging the gap between non-uniform and uniform quantization, GSTE [21] leverages a generalized straight-through estimator to optimize input thresholds, allowing for more flexible and accurate quantization while maintaining hardware efficiency. Moreover, PowerQuant [22] proposes a data-free non-uniform quantization method by optimizing the exponent parameter of a continuous automorphism function (i.e., a power function). Specifically, it searches for the optimal quantization operation while preserving mathematical properties in DNNs, such as matrix multiplication. By minimizing quantization reconstruction error, PowerQuant determines the best exponent parameter for each layer to closely fit the original weight distribution. Overall, these approaches are particularly advantageous as they enable the quantization process to adapt to the empirical distribution of weights, thereby minimizing quantization errors.

A representative non-uniform quantization method is Reorder-based Post-training Quantization (RPTQ) [23], which extends the framework of OPTQ by incorporating K-means clustering across weight matrix channels. This channel-wise clustering allows for a more granular and adaptive quantization scheme, tailored to the specific characteristics of each channel. On this basis, SqueezeLLM [24] introduces a sophisticated approach that reduces quantization error by using sparse matrices to store outliers and sensitive values separately. Furthermore, it employs the Fisher information matrix as an approximation of the Hessian matrix to quantify weight sensitivity. Subsequently, a sensitivity-weighted K-means clustering algorithm is utilized for the quantization process. The Fisher information matrix, defined as the negative expectation of the Hessian matrix of the log-likelihood function, provides a computationally efficient alternative to the Hessian matrix. Substituting the Fisher information matrix into Formula Eq. (3) yields:

$$Loss(w + \Delta) - Loss(w) \approx \frac{1}{2} \Delta^T \text{diag}(F) \Delta, \quad (4)$$

The Fisher information matrix offers a computational advantage over the Hessian matrix by involving only first-order derivatives of the log-likelihood function, as opposed to the second-order derivatives required by the latter. This reduced computational complexity renders the Fisher information matrix a more practical choice for large-scale models.

3. Methods

Despite the substantial advancements achieved in the domain of LLM quantization, several critical challenges persist, particularly in the realm of ultra-low-bit quantization. A prime example is the application of 2-bit quantization, a scenario visually represented in Fig. 2, which illustrates the inherent difficulties associated with such extreme compression levels.

This paper is dedicated to further refining and optimizing low-bit quantization methodologies for LLMs. Specifically, we address the aforementioned challenges through the introduction of Sensitivity-based error minimization strategy and non-uniform Quantization. Our proposed approach dynamically allocates quantization points based on the intrinsic sensitivity of individual weights. This strategy prioritizes the quantization errors of highly sensitive weights, thereby suppressing quantization errors under extreme quantization conditions. The algorithm flow

is shown in Fig. 3. Subsequents of the section will provide a comprehensive and detailed exposition of our proposed methodology.

3.1. Outliers and sensitive values

Outliers. As shown in Fig. 4, the frequency distribution of weight values from various layers of a large language model (LLM) is depicted. Each subplot, labeled from (a) to (p), presents a histogram of weights with a fitted normal distribution curve. The histograms exhibit a bell-shaped pattern, indicating that the weight distributions closely resemble a normal distribution. This is evident from the concentration of weights around the mean and the symmetric shape of the curves. Given this observation, we derive an outlier threshold based on normal distribution quantiles. Employing the Z-score, a measure of data point deviation from the mean, we establish a threshold for outlier identification, that is, utilizing a Z-score of 3, consistent with the 68-95-99.7 rule. This threshold effectively isolates approximately 0.27 % of extreme values as outliers, facilitating robust quantization. The mathematical formulation underpinning this outlier detection is as follows:

$$w_{\text{outlier}} = \{w < \mu - 3\sigma \text{ or } w > \mu + 3\sigma\} \quad (5)$$

where μ is the mean and σ is the standard deviation. The result is also shown in Fig. 4, a histogram of 16 randomly selected rows of weight values from the first layer of the LLaMa-7B model, fitted with a normal distribution (red line), shows that most weights are concentrated around the mean.

It is worth noting that the actual distribution of weights in large language models (LLMs) does not typically conform to a single bell-shaped (unimodal Gaussian) distribution. In fact, it may exhibit characteristics of a multimodal distribution, with multiple peaks. To illustrate this, we randomly selected a linear layer from LLaMa-7B and visualized its weight distribution using two different methods. In the first method (a), we directly plotted the histogram of all weights. In the second method (b), we separated the weights into two groups—those greater than zero and those less than or equal to zero—and plotted their histograms independently before concatenating them for visualization. The result is shown in Fig. 5, the two histograms exhibit noticeable differences in shape. Method (a) yields a relatively smoother and more symmetric curve, while method (b) reveals an asymmetric and possibly multimodal pattern, suggesting that the weights are not symmetrically distributed around zero. Despite this apparent discrepancy in histogram shape, the impact on the calculation of outliers appears to be minimal. When applying a standard Z-score-based thresholding method with a cutoff of $\alpha = 3$, the resulting outlier bounds are comparable: method (a) produces lower and upper bounds of -0.0675 and 0.0668, respectively, while method (b) yields -0.0729 and 0.0660. This observation suggests that although the empirical distribution of LLM weights may deviate from the idealized Gaussian assumption, such deviations do not significantly affect the identification of outliers in practice. This can also be intuitively inferred from the histograms: although the overall shapes differ, the underlying distributional characteristics (such as the range and density of the majority of weights) remain largely consistent across both visualizations.

Sensitivities. Drawing inspiration from SqueezeLLM [24], we employ the Fisher information matrix, an equivalent representation of the Hessian matrix, to quantify the sensitivity of individual weights. Furthermore, we adopt quantile-based thresholds to identify a subset of critical, highly sensitive weights.

Following the extraction of these sensitive values and outlier weights, we preserve them in full precision. Given that the cardinality of these weights constitutes a negligible fraction of the total weight parameters (less than 1 %), their separate storage does not substantially impact the overall model size reduction achieved through compression. This approach effectively mitigates the performance degradation typically associated with extreme low bit quantization, while maintaining a compact model footprint.

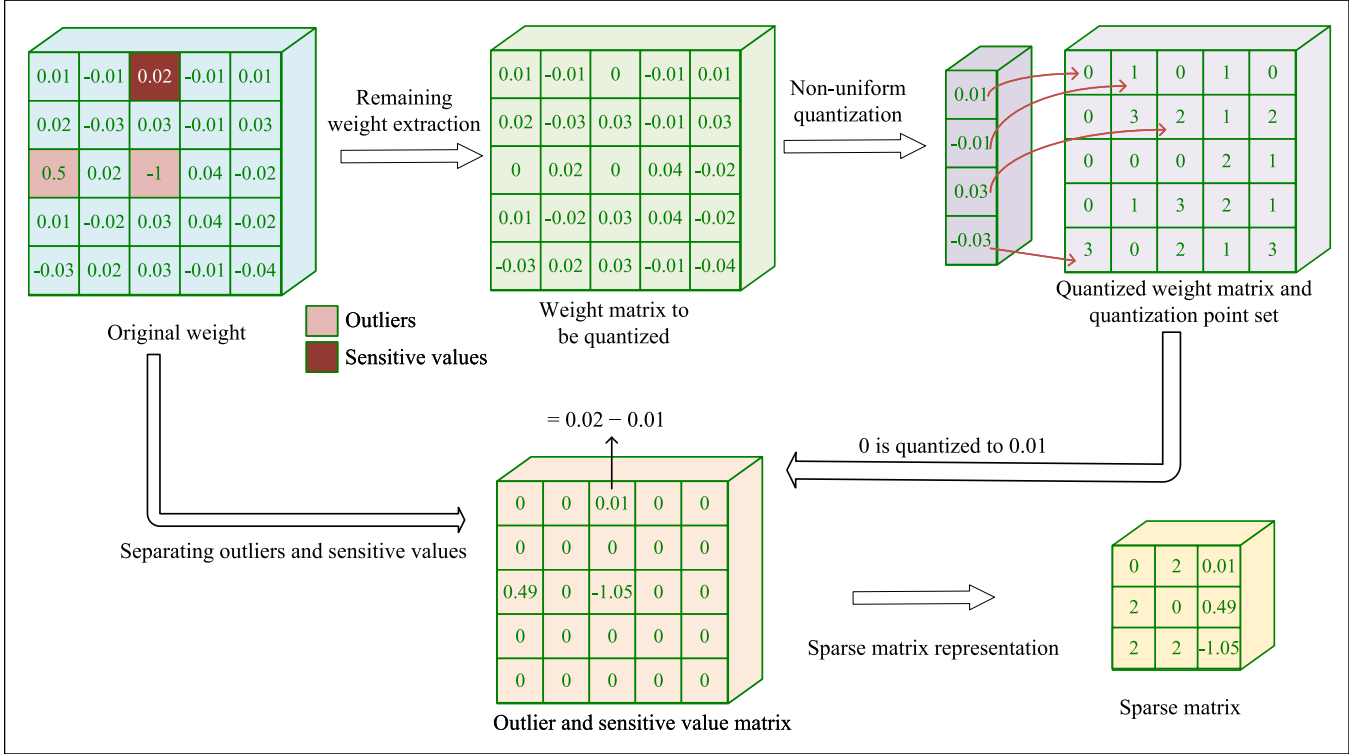


Fig. 3. The diagram of the proposed SEMQ. The algorithm first extracts the outliers and sensitive values in the weights, and then performs non-uniform quantization based on sensitivity on the remaining parts to find the best quantization solution. The proposed model employs a hybrid quantization scheme wherein quantized weights are the addition of non-uniformly quantized weight values and a sparse matrix containing outlier and sensitivity-critical data points. To mitigate the potential for zero-offset interference on these retained outliers and sensitive points, a correction factor (e.g., 0.01 as shown in the figure), a correspondence to the quantized value that represents zero, is subtracted from each element of the sparse matrix, thereby reducing quantization-induced error.

3.2. Non-uniform quantization based on sensitivity-based error minimization

Building upon the established definition of weight sensitivity presented in the previous section, we derive the sensitivity metric for each weight parameter using backpropagation. To minimize the adverse effects of quantization-induced perturbations on model accuracy, we propose a strategy that aligns sensitive weights more closely with the quantized values. This approach can be equivalently formulated as the minimization of a quantization error, which is formally defined as follows:

$$loss = \sum_i (W_i - \tilde{W}_i) \times S_i, \quad (6)$$

where W_i denotes the original i -th weight value, $\tilde{W}_i$ is the corresponding quantized value, and S_i is the sensitivity of the weight i . The quantization error can be minimized by adjusting the quantized value $\tilde{W}_i$.

The process of weight quantization inherently influences the correspondence between pre- and post-quantization weights. This interdependence arises from the principle that each weight is assigned to the quantized value that minimizes its distance within the set of all quantization points. In essence, the determination of quantization points for a weight matrix simultaneously establishes the mapping between the original and quantized weights.

Despite the development of various estimation techniques, accurately identifying an optimal set of quantization points remains a significant challenge. However, given an initial approximation of quantization points, it becomes feasible to refine these points iteratively. For each quantization point and its associated set of pre-quantization weights, we can compute a more refined quantization point value, as illustrated

Algorithm 1 Sensitivity-based error minimization quantization.

Input: Weight value matrix W , weight sensitivity matrix S , quantization bit n , Z-score z

Output: Quantized weight matrix W_q , quantization point set V

- 1: $\mu, \sigma \leftarrow fit(W)$ ▷ Fitting the mean and standard deviation
- 2: $V \leftarrow init(\mu, \sigma, n, z)$ ▷ Initialization of Quantization Point
- 3: $W_q \leftarrow getNearest(W, V)$ ▷ For each value of W , find the nearest value in V
- 4: $loss \leftarrow \sum_i abs(W_i - V_{\tilde{W}_i}) \times S_i$
- 5: **while** $loss$ is not converged **do**
- 6: $V \leftarrow update(V, W, S)$
- 7: $W_q \leftarrow getNearest(W, V)$
- 8: $loss \leftarrow \sum_i abs(W_i - V_{\tilde{W}_i}) \times S_i$
- 9: **end while**
- 10: **return** W_q, V

by the following formula:

$$V = \frac{\sum_i V_{W_i} \times S_i}{\sum_i S_i}, \quad (7)$$

where W_i denotes the original i -th weight value, and S_i is the sensitivity of the weight i . The refined quantization point, denoted as V , is computed by determining the weighted average of all corresponding pre-quantization weights. As previously established, the confirmation of a quantization point inherently defines the mapping between the original weight values and the quantized representations. Consequently, this refinement process can be executed iteratively, resulting in a progressive reduction of the quantization loss until convergence is achieved. The complete algorithmic procedure is detailed in Algorithm 1.

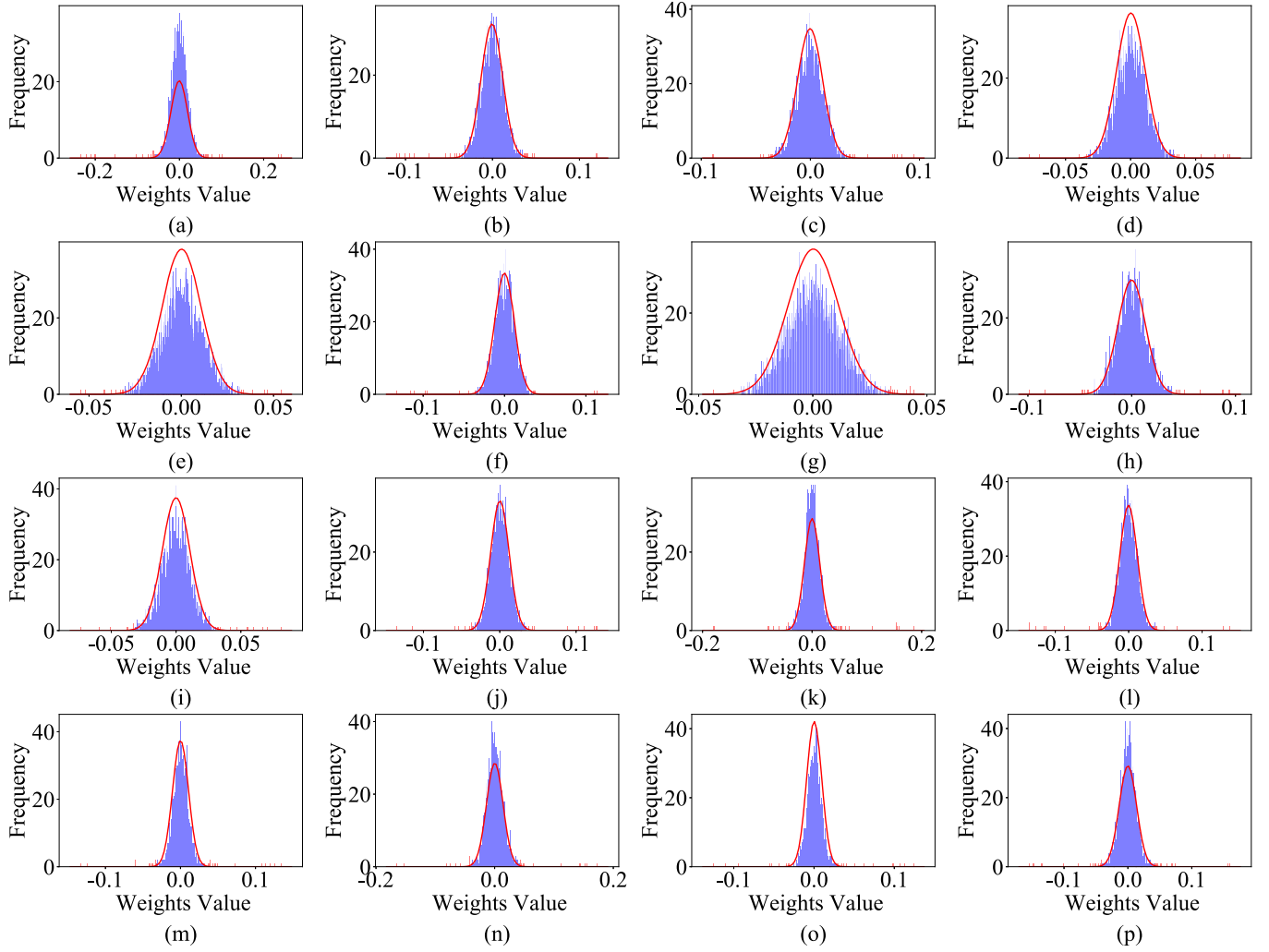


Fig. 4. LLM linear weight distribution histogram, where the red curve is the normal distribution curve fitted according to the mean and standard deviation of the weights, and the red ones (boxes) are outliers. We randomly selected 12 rows on the linear weights of the first layer of the LLaMA-7B model to draw this graph.

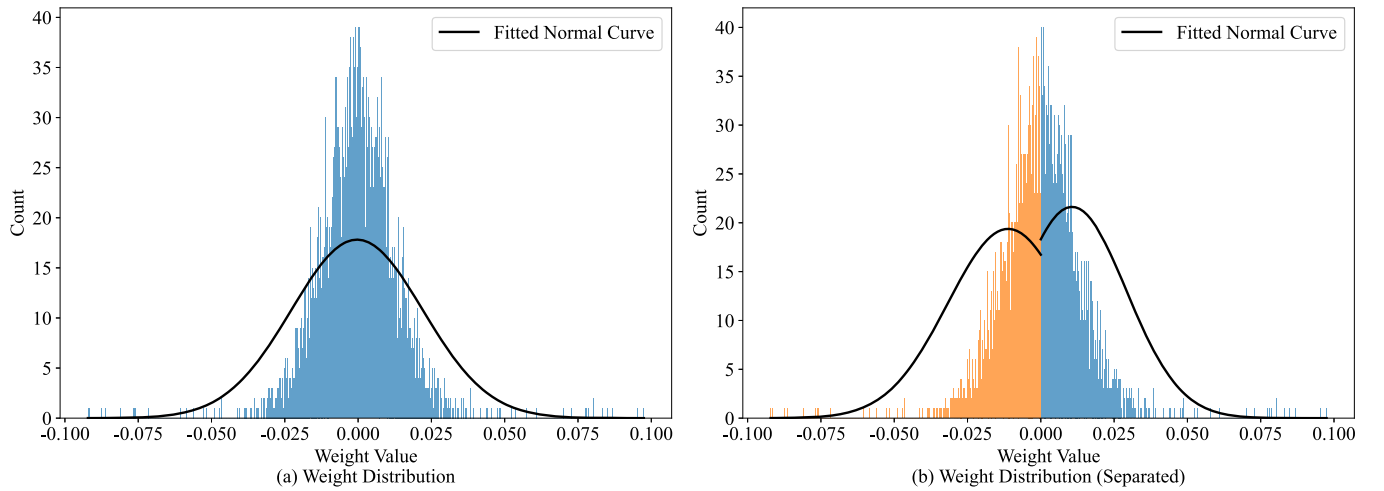


Fig. 5. Histogram of the weight distribution from the first-layer Q linear matrix in LLaMA-7B. (a) shows the histogram plotted using all weights directly. (b) presents the histogram by separately plotting the weights greater than 0 and those less than or equal to 0, and then merging them into a single figure.

The determination of initial quantization points can be effectively guided by the inherent bell-shaped distribution characteristics of the weight parameters. Specifically, we leverage the Z-score metric to compute these initial points, as detailed in [Algorithm 2](#). This computational approach is strategically designed to preclude the selection of quantization points that coincide with the central tendency and extreme boundaries of the normal distribution (i.e., $\mu \pm 3\sigma$). Such coincidences may negatively affect the convergence behavior of the loss function in subsequent optimization iterations.

Algorithm 2 Initialization of quantization point.

Input: Weight value matrix W , quantization bit n , Z-score z
Output: Quantization point set V

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1:  $V \leftarrow$  Empty set
2:  $\mu, \sigma \leftarrow fit(W)$  ▷ Fitting the mean and standard deviation
3: for  $i = 1$  to  $2^{n-1}$  do
4:    $V \leftarrow V \cup \left\{ \mu + \frac{i}{2^{n-1}+1} z\sigma \right\}$ 
5:    $V \leftarrow V \cup \left\{ \mu - \frac{i}{2^{n-1}+1} z\sigma \right\}$ 
6: end for
7: return  $V$ 

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4. Experiments

4.1. Experimental setups

This research undertakes an empirical investigation into the efficacy of ultra-low bit quantization techniques for Large Language Models (LLMs), building upon the foundational implementation provided by the SqueezeLLM [24] repository. Specially, SqueezeLLM introduces a PyTorch [25] framework for the 3-bit and 4-bit quantization of LLMs, encompassing a CUDA-optimized implementation of compressed matrix storage to mitigate the computational overhead associated with quantized representations. We extend this framework by implementing and evaluating a 2-bit quantization scheme, thereby pushing the boundaries of model compression.

Consistent with the methodology employed by SqueezeLLM, our quantization process necessitates the computation of weight sensitivity metrics. This computation relies on the C4 [26] as a calibration dataset. It is crucial to emphasize that the C4 dataset utilized for calibration is distinct from the evaluation datasets employed in subsequent performance assessments. We employ a dedicated calibration dataset that is representative of the general language domain, ensuring that both the calibration process and the resulting quantized models remain task-agnostic. This preserves the intrinsic properties of the quantization method and reduces performance deviations across different downstream tasks. As to the metric, Perplexity (PPL), a widely recognized metric for evaluating language model performance, is employed to quantify the accuracy of the quantized models. Lower PPL values indicate superior performance, reflecting a closer approximation to the original, unquantized model's behavior.

Our experimental framework primarily utilized models from the LLaMA [1,2] and OPT [27] families. All experiments, encompassing both quantization and evaluation procedures, were conducted on an NVIDIA A800 GPU with 80 GiB of memory. To maintain experimental consistency and control, all processes were executed on a single GPU device. Our quantization pipeline largely mirrors the sequential steps outlined in SqueezeLLM [24], with a notable optimization: the integration of several procedural phases. Specifically, we consolidated the following steps into a streamlined process: loading the pre-trained LLMs into GPU memory, creating a copy of the model and converting its weights to float16 precision, computing the Fisher information matrix using the calibration dataset, performing non-uniform quantization based on the computed sensitivity metrics, and serializing the quantized model for persistent storage. This consolidation enhances the efficiency of the quantization process and reduces computational overhead.

4.2. Results on 2-bit quantization

A comprehensive evaluation of various quantization methodologies was performed, focusing on extreme low bit quantization levels. For our evaluation, we tested a variety of quantization methods including Round-to-Nearest (RTN) quantization, GPTQ [11], AWQ [12], OmniQuant [18], PB-LLM [28], BWA [29], QuaRot [30] and Atom [31]. These methods represent the current state-of-the-art in ultra-low bit quantization and were evaluated under 2-bit quantization scenarios. To provide a rigorous and comprehensive assessment of language generation capabilities, we employed the C4 [26] and WikiText-2 [32] datasets, which are widely recognized benchmarks for evaluating language model performance.

The empirical evaluation of our proposed 2-bit quantization methodology reveals a substantial performance advantage over contemporary state-of-the-art techniques, as detailed in [Table 1](#). It's evident that SEMQ outperforms other methods in terms of perplexity (PPL) on both LLaMA-7B and LLaMA-13B models. For example, on LLaMA-7B, SEMQ achieves PPLs of 9.83 and 11.62 on WikiText-2 and C4, respectively, significantly better than PB-LLM (102.36 and 100.38). On LLaMA-13B, SEMQ's PPLs are 7.73 and 9.43, again outperforming all others. SEMQ also maintains competitive memory usage. It uses 2.4 GiB for LLaMA-7B and 4.2 GiB for LLaMA-13B, similar to other methods but with superior performance. In contrast, GPTQ, AWQ, and RTN show much higher PPL values, indicating poorer performance under 2-bit quantization. Furthermore, it is crucial to note that the 'Block size' parameter, which dictates the granularity of quantization, varies across methods. Within the same method, a smaller block size means that the quantized weights need to cover fewer pre-quantization weights, resulting in smaller quantization errors. SEMQ, similar to other methods that do not explicitly define a block size, performs quantization at the weight channel (row) level. In such cases, the effective block size is naturally determined by the structure of the linear layers in LLMs of varying sizes.

4.3. Comparison on efficiency

To further substantiate the efficacy of our method, we conducted a comparative analysis of 3-bit quantization performance, leveraging the SqueezeLLM [24] repository as a baseline, given that our methodology is built upon its foundational implementation. The results of this comparison are presented in [Table 2](#). Our proposed method, SEMQ, consistently achieves the best performance across all evaluated techniques under 3-bit quantization. For instance, on the WikiText2 dataset with LLaMA-7B model, SEMQ achieves a perplexity of 6.13, which is similar to SqueezeLLM (6.13) and APTQ [33] (6.24) but significantly better than other methods like RTN (25.61) and GPTQ (7.55). Similarly, on the C4 dataset, SEMQ achieves a perplexity of 7.54, outperforming RTN (25.61) and GPTQ (9.55). The memory footprint reduction achieved by SEMQ is also noteworthy. Our method compresses the LLaMA-7B model to 3.1 GiB, which is a substantial reduction from the baseline size of 12.7 GiB. This compression ratio facilitates deployment in resource constrained environments without incurring substantial performance penalties. We also compared the inference speed under 3-bit quantization. The result is shown on [Tables 2](#) and [3](#). Benefiting from the reduced model size, the quantized models achieved significant acceleration compared to the baseline. However, most methods exhibited similar speedups, as quantization, unlike pruning, does not eliminate a large amount of redundant computation.

To further validate the broad applicability of our method, we also conducted 3-bit quantization experiments on the OPT model ([Table 3](#)). The results demonstrate that our approach is not only effective for the LLaMA model but also exhibits practical utility on other large-scale models.

Our experimental results demonstrate the efficiency and effectiveness of SEMQ in 2-bit and 3-bit quantization. While the performance

Table 1

The performance of 2-bit quantization on different datasets on the LLaMA-7B and LLaMA-13B model. All method parameters are set to default, and the baseline is float16. We added a column to indicate the memory usage when the model uses float32(full-precision) for inference, but in general LLMs use float16/bfloat16 for inference, which is almost lossless.

Method	Block Size	LLaMA-7B			LLaMA-13B		
		Mem.(GiB↓)	WikiText2(PPL↓)	C4(PPL↓)	Mem.(GiB↓)	WikiText2(PPL↓)	C4(PPL↓)
Baseline	-	12.7	5.68	7.08	24.3	5.09	6.61
Baseline(full-precision)	-	25.9	-	-	51.4	-	-
GPTQ	-	2.1	2.1e3	689.13	3.8	5.5e3	2.5e3
GPTQ	64	2.2	22.10	17.71	3.9	10.06	11.70
AWQ	128	2.1	2.6e5	1.9e5	3.8	2.8e5	2.3e5
AWQ	64	2.2	2.5e5	2.8e5	3.9	2.7e5	2.2e5
RTN	-	2.1	1.1e5	1.3e5	3.8	6.8e4	5.6e4
OmniQuant	-	2.2	15.47	24.89	4.0	13.21	18.31
PB-LLM	128	-	102.36	100.38	-	36.60	39.51
BWA	128	2.7	8.58	12.27	4.8	7.19	10.18
QuaRot	-	-	14.39	27.70	-	11.14	20.80
Atom	-	-	16.65	33.87	-	11.69	19.55
SEMQ (Ours)	-	2.4	9.83	11.62	4.2	7.73	9.43

Table 2

The performance of 3-bit quantization on different datasets on the LLaMA-7B and LLaMA-13B model. SqueezeLLM(0.85 %) means that the extracted outliers and sensitive values account for 0.85 % of the total number of parameters. Otherwise, all method parameters are set to default, and the baseline is float16.

Method	LLaMA-7B				LLaMA-13B			
	Mem. (GiB↓)	Speedup (↑)	WikiText2 (PPL↓)	C4 (PPL↓)	Mem. (GiB↓)	Speedup (↑)	WikiText2 (PPL↓)	C4 (PPL↓)
Baseline	12.7	1.0 ×	7.08	5.68	24.3	1.0 ×	5.09	6.61
RTN	2.9	2.3 ×	25.61	25.61	5.3	2.7 ×	11.78	13.24
GPTQ	2.9	2.3 ×	7.55	9.55	5.3	2.7 ×	6.22	8.22
SqueezeLLM	2.9	2.1 ×	6.32	7.75	5.4	2.4 ×	5.60	7.08
SqueezeLLM(0.85 %)	3.1	1.9 ×	6.13	7.56	5.8	2.2 ×	5.45	6.92
APTQ	3.1	-	6.24	6.76	-	-	5.56	7.01
SEMQ (Ours)	3.1	1.9 ×	6.13	7.54	5.9	2.2 ×	5.42	6.91

Table 3

The performance of 3-bit quantization on different datasets on the OPT-6.7B and OPT-13B model. SqueezeLLM(0.5 %) means that the extracted outliers and sensitive values account for 0.5 % of the total number of parameters. AWQ(g128) means that AWQ method with block size 128. Otherwise, all method parameters are set to default, and the baseline is float16. “div” means that the perplexity is diverged.

Method	OPT-6.7B				OPT-13B			
	Mem. (GiB↓)	Speedup (↑)	WikiText2 (PPL↓)	C4 (PPL↓)	Mem. (GiB↓)	Speedup (↑)	WikiText2 (PPL↓)	C4 (PPL↓)
Baseline	12.4	1.0 ×	10.86	11.74	23.9	1.0 ×	10.12	11.20
RTN	3.4	-	div.	div.	5.6	-	div.	div.
AWQ (g128)	3.6	2.0 ×	11.41	12.31	6.4	2.3 ×	10.67	12.61
SqueezeLLM	3.3	2.2 ×	11.70	12.44	5.8	2.5 ×	11.76	12.69
SqueezeLLM(0.5 %)	3.6	2.0 ×	11.31	12.18	6.4	2.3 ×	10.54	11.57
APTQ	-	-	11.52	12.38	-	-	10.73	12.01
SEMQ (Ours)	3.4	2.1 ×	11.39	12.14	5.9	2.4 ×	10.53	11.52

gains are less dramatic compared to 2-bit quantization levels, SEMQ consistently outperforms other state-of-the-art methods, achieving a favorable balance between model compression and performance preservation.

4.4. The effect of generalization

Subsequently, we conducted a comprehensive evaluation of our proposed methodology across a spectrum of LLMs with varying parameter sizes within the OPT family. The performance metrics obtained from these experiments are detailed in Fig. 6, which illustrates the scalability and effectiveness of our approach.

As shown in Fig. 6, SEMQ consistently demonstrates superior performance across different model sizes. The solid lines represent the performance of the SEMQ 2-bit quantized models on the C4 [26] and WikiText2 [32] datasets, while the dashed lines indicate the performance of

the original models(float16). Across all evaluated model scales (ranging from 1.3 billion to 13 billion parameters), SEMQ consistently exhibits only minimal losses in perplexity (PPL) compared to the baseline, indicating that our method is effective across various sizes of LLMs. For instance, on the C4 dataset, the PPL of SEMQ remains only slightly higher than that of the original model regardless of the parameter scale, demonstrating that the quantization performance does not significantly degrade even for smaller models. A similar trend is observed on the WikiText2 dataset.

These results highlight the robustness and generalizability of SEMQ. The method’s ability to maintain efficacy across diverse LLM scales underscores its broad applicability. This finding suggests that SEMQ is not limited to a specific model size or architecture but offers consistent performance enhancement across a range of LLM configurations. Thus, SEMQ provides a reliable solution for efficient quantization of large language models.

Table 4

The zero-shot experiments were performed on LLaMA-7B and LLaMA-13B, using 2-bit quantization. All methods used default settings, and the baseline is float16.

Model	Bits	Method	Accuracy(% ↑)					
			ARC-e	ARC-c	PIQA	HellaSwag	Winogrande	Avg.
LLaMA-7B	16	Baseline	52.53	41.38	77.37	72.99	66.85	62.22
	2	GPTQ	26.60	29.50	52.80	26.30	49.30	36.90
	2	AWQ	25.76	29.44	50.05	25.93	49.96	36.23
	2	OmniQuant	44.49	29.69	68.66	54.32	55.56	50.54
	1.7	PB-LLM	28.20	24.60	54.60	28.70	50.60	37.34
	2	DB-LLM	44.70	33.62	72.14	60.71	61.01	54.44
	2	BWA	45.03	30.89	68.88	55.41	59.35	51.91
	2	QuaRot	37.88	26.62	59.92	41.56	54.62	44.12
	2	Atom	35.23	26.11	57.24	36.77	50.51	41.17
	2	SEMQ (Ours)	42.63	29.18	70.46	59.93	58.88	52.22
LLaMA-13B	16	Baseline	59.85	44.62	79.05	76.22	70.09	65.97
	2	GPTQ	49.58	36.01	71.44	63.34	62.43	56.56
	2	AWQ	27.19	28.92	50.76	26.29	47.91	36.21
	2	OmniQuant	49.54	33.70	73.01	62.10	61.96	56.06
	1.7	PB-LLM	40.99	28.33	62.89	40.77	58.09	46.21
	2	DB-LLM	51.18	37.54	74.16	68.29	64.72	59.18
	2	BWA	49.92	36.26	71.98	60.52	61.80	56.10
	2	QuaRot	34.60	24.23	59.09	35.03	51.38	40.87
	2	Atom	40.36	29.52	61.15	45.62	51.22	45.57
	2	SEMQ (Ours)	47.20	35.23	74.43	70.83	67.09	58.96

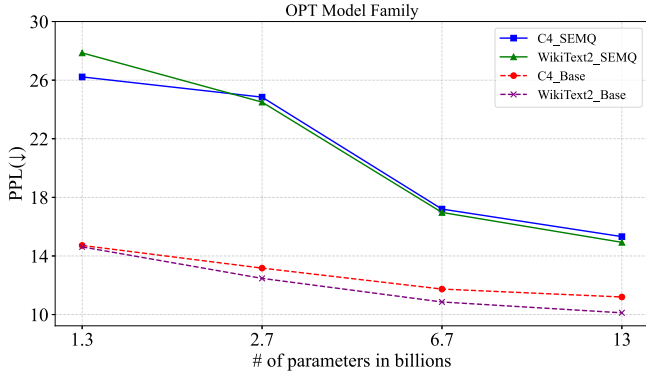


Fig. 6. SEMQ 2-bit quantization performance on the OPT family. The dashed line connects the performance of the base model(bfloat16) on the corresponding dataset. The solid line connects the performance of the SEMQ 2-bit quantized model on the corresponding data set. These values represent PPL, and the lower the better.

4.5. Results on zero-shot tasks

To quantify the potential accuracy degradation resulting from the quantization process, we conducted zero-shot task evaluations. These evaluations were performed on the LLaMA-7B and LLaMA-13B models, quantized to 2-bit precision. We utilized the language model evaluation harness[34] to ensure standardized and reproducible evaluation procedures. The baseline methods considered for comparative analysis included GPTQ [11], AWQ [12], OmniQuant [18], PB-LLM [28], BWA [29], QuaRot [30] and Atom [31]. We strategically selected a diverse set of datasets to comprehensively assess the models' capabilities across various cognitive tasks. These datasets include the AI2 Reasoning Challenge (ARC) [35], which evaluates scientific reasoning in primary school level questions; Physical Interaction: Question Answering (PIQA) [36], which assesses common-sense physical knowledge; HellaSwag [37], which tests common-sense reasoning in everyday scenarios; and Winogrande [38], a large-scale common-sense reasoning benchmark. These datasets collectively provide a robust evaluation framework, enabling a detailed analysis of the impact of ultra-low bit quantization on LLM performance across diverse cognitive tasks.

The efficacy of our proposed methodology was rigorously evaluated through a series of zero-shot learning experiments, the results of which are meticulously documented in Table 4. Within this table, the presented numerical values represent the classification accuracy achieved by the respective models. It is imperative to note that in this context, higher numerical values directly correlate with superior model performance, indicating a greater ability to accurately classify unseen data. In the case of LLaMA-7B, SEMQ demonstrated superior performance compared to most other methods. Most of other methods like GPTQ (36.9) and AWQ (36.228) showed substantial performance degradation under 2-bit quantization. This indicates that SEMQ effectively preserves critical model information even under extreme compression, maintaining the model's ability to perform complex reasoning tasks. Similarly, on the LLaMA-13B model, SEMQ outperforms most other methods, achieving an average accuracy of 58.96. The performance gap compared to the best-performing method, DB-LLM (with an average accuracy of 59.18), is minimal. It is worth noting that DB-LLM adopts a fixed group size of 64, whereas SEMQ determines the group size based on the dimensionality of model weights (4096 for most layers in LLaMA-7B). The performance gap between SEMQ and other methods, such as GPTQ (56.56) and AWQ (36.214), further underscores the robustness and effectiveness of our proposed method.

The results from these zero-shot task evaluations highlight the superiority of SEMQ in preserving the performance of LLMs under 2-bit quantization. By retaining the most salient weights, SEMQ minimizes information loss and maintains the model's ability to understand, reason, and generate coherent responses. This is crucial for applications requiring semantic understanding and generative capabilities, such as natural language processing, question answering, and text summarization.

4.6. Ablation studies

To rigorously assess the contribution of individual components within our proposed Sensitive and Outlier-aware Mixed-precision Quantization (SEMQ) framework, we conducted a comprehensive ablation study. The results of this study are presented in Table 5, which evaluates the impact of outlier extraction on the performance of SEMQ under both 2-bit and 3-bit quantization for LLaMA-7B and LLaMA-13B models. The experimental findings clearly demonstrate the critical role of outlier extraction in achieving effective and accurate weight quantization.

Table 5

Ablation studies on 2-bit and 3-bit quantization on different datasets for the LLaMA-7B and LLaMA-13B models. The values represent PPL, with lower values indicating better performance.

Config	LLaMA-7B				LLaMA-13B			
	2 bit		3 bit		2 bit		3 bit	
	C4	WikiText2	C4	WikiText2	C4	WikiText2	C4	WikiText2
W/o outlier extraction	16.48	16.15	7.80	6.32	13.87	12.09	7.09	5.55
W outlier extraction	11.62	9.83	7.54	6.13	9.43	7.73	6.91	5.42

Table 6

Ablation studies of the initial quantization point selection method on 2-bit and 3-bit quantization of LLaMA-7B and LLaMA-13B models. The value represents the number of iterations, and the lower the value, the faster the method. We selected 1 column of data in the linear matrix of the first layer of the model as the sample weight for the study.

Config	LLaMA-7B		LLaMA-13B	
	2 bit	3 bit	2 bit	3 bit
Based on quantile	21	21	37	38
Based on random selection	32	32	47	47

Specifically, when outliers are not extracted and treated separately, the performance of SEMQ under 2-bit quantization is significantly compromised. As shown in Table 5, without outlier extraction, the perplexity (PPL) values for LLaMA-7B on C4 [26] and WikiText2 [32] datasets increase to 16.48 and 16.15, respectively, compared to 11.62 and 9.83 with outlier extraction. Similarly, for LLaMA-13B, the PPL values rise to 13.87 and 12.09 without outlier extraction, versus 9.43 and 7.73 when outliers are extracted.

This phenomenon is due to the characteristics of row-wise weight quantization. Without mitigating outliers, quantization points are disproportionately influenced by extreme values, skewing them towards the tails of the distribution instead of the central tendency. This results in higher quantization error as most weights are inadequately represented. The effect is more pronounced under 2-bit quantization due to the limited number of levels. In contrast, outlier extraction shows benefits even under 3-bit quantization. For example, the PPL values for LLaMA-7B on C4 and WikiText2 datasets decrease from 7.80 / 6.32 to 7.54 / 6.13, and for LLaMA-13B, they drop from 7.09 / 5.55 to 6.91 / 5.42 when outliers are extracted. These results highlight the importance of addressing outliers in various bit-widths.

Our method also incorporates a quantile-based initialization strategy for selecting initial quantization points. This approach aims to accelerate the iterative optimization process by providing a better starting point for quantized values. We compare it with the commonly used random initialization in K-Means, as shown in Table 6. The results demonstrate that the quantile-based initialization offers a noticeable improvement in convergence speed.

Overall, the ablation study emphasizes the significance of handling outliers during quantization. By mitigating their influence, SEMQ achieves more accurate and efficient weight quantization, improving model performance across different datasets and bit-widths.

5. Conclusion

In this study, we introduce Sensitivity-based Error Minimization Quantization (SEMQ), a novel low-bit compression methodology designed for LLMs. The proposed SEMQ combines non-uniform quantization with a sensitivity-based iterative error minimization approach to reduce quantization-induced losses, preserving the model's representational capacity. Our systematic comparison against state-of-the-art quantization methods highlights SEMQ's superior performance in achieving ultra-low-bit (2-bit) quantization while maintaining high per-

formance fidelity. The results of our empirical analysis demonstrate SEMQ's effectiveness and robustness, underscoring its potential to advance real-world applications in scenarios where computational and memory resources are limited. We believe this breakthrough enables efficient deployment of LLMs on resource-constrained platforms, including edge devices, mobile computing systems, and IoT applications.

Limitations and future work: While our method advances the state-of-the-art in 2-bit quantization of LLMs, it is not without limitations. The computation of weight sensitivity is reliant on a calibration dataset, potentially introducing bias. Furthermore, the convergence criterion for the iterative method used to mitigate quantization errors requires further investigation to ensure robustness. These limitations highlight important directions for future research. Developing alternative methods for computing weight sensitivity that are less reliant on specific datasets could reduce potential biases. Refining the convergence criteria for the iterative error mitigation process could lead to more effective and efficient solutions. We anticipate that our work will help inspire future research into low-bit quantization of LLMs, facilitating broader deployment in various applications.

CRedit authorship contribution statement

Dongmin Li: Writing – original draft, Software, Methodology; **Xiurui Xie:** Writing – review & editing, Methodology, Conceptualization; **Dongyang Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition; **Athanasios V. Vasilakos:** Methodology, Data curation; **Man-Fai Leung:** Validation, Resources, Formal analysis.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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